

Week 05. Hypothesis testing II. Central Limit Theorem. T-distribution. Confidence intervals. t-test

Part 1: Statistics and Econometrics. MK68-2024/25

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Content

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- Normally distributed population and the confidence intervals
- The Central Limit Theorem (CLT)
- The Confidence Interval
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t-test with the Conditional Cash Transfers (CCTs)

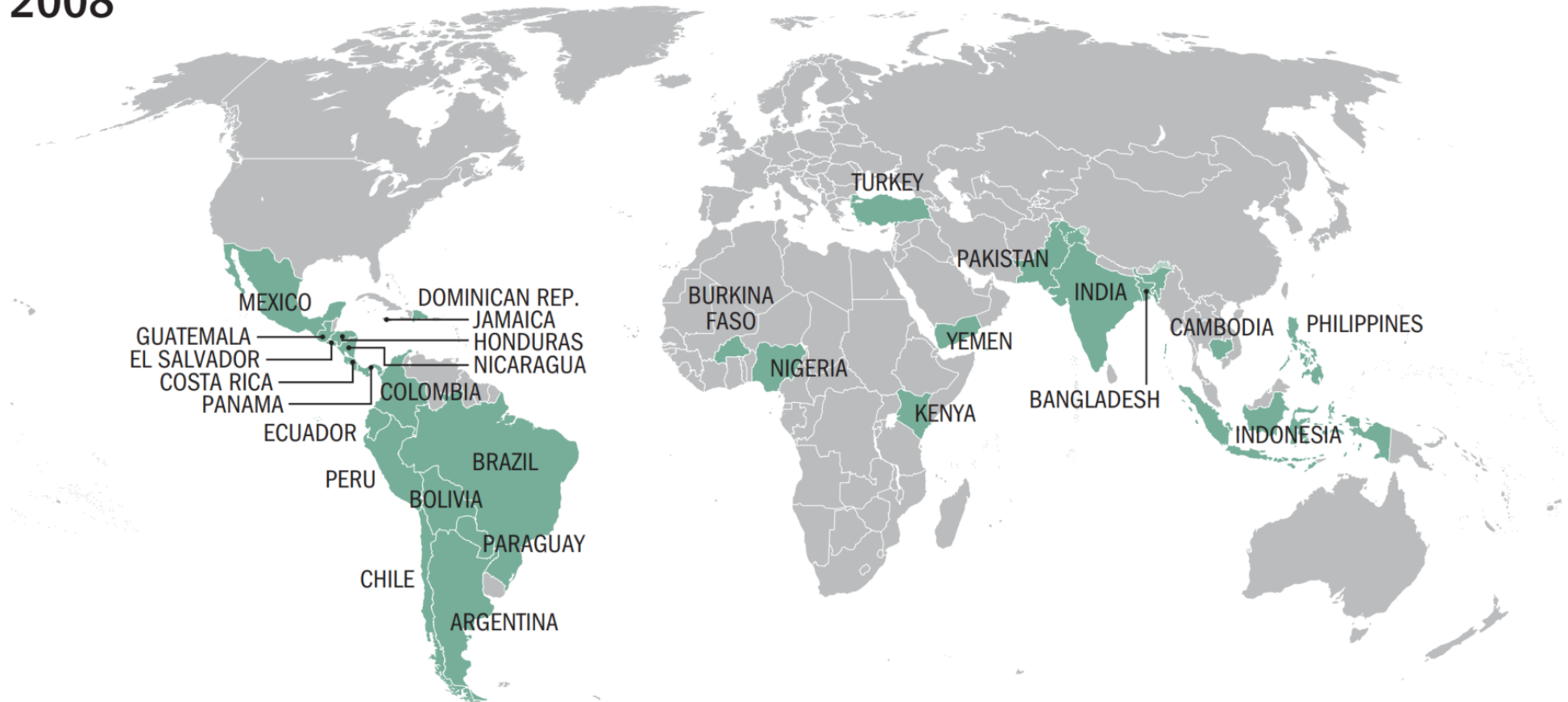
Let us consider an example of CONDITIONAL CASH TRANSFERS' causal effect on children school enrollment.

See: CH 5. The Impact of CCT Programs on the Accumulation of Human Capital from ([Fiszbein et al. 2009](#)).

Conditional Cash Transfers (CCTs)

CCTs are programs that **transfer cash** on the **condition** of making pre-specified investments in **human capital and children**. ([Fiszbein et al. 2009, fig. 1](#))

2008



Cause and Effect

1. Cause / Treatment:

- CCTs (monetary incentive) + Pre-specified conditions

2.a Outcome / Factual

- School enrollment
- in treatment;

2.b Counterfactual

- School enrollment
- in control;

3. ATE: Effect = Factual – Counterfactual

- Average Treatment Effect = Enrollment Rate Difference

What makes “Difference in Enrollment Rate” a causal effect?

What statistical concepts ensure that the difference in school enrollment between the treatment and control is caused by the treatment?

1. Research Designed with the RCT:

- Randomization between the Control and Treatment group;

2. the Law of Large Numbers (LLN):

- Both random groups are equal on average;

3. The RCT + LLN = **Ceteris Paribus**;

- All other things are equal between the Control and Treatment groups;

Good controls in the RCT

The control group is a perfect **counterfactual** when **three conditions are met**:

1. has the same characteristics, on average, as the treatment group;
2. remains unaffected by the treatment; and
3. reacts to the treatment in the same way as the treatment group;

When these three conditions are met, the treatment causes the effect ceteris paribus ([Gertler et al. 2016](#)).

How to check that ceteris paribus is ensured?

What statistical tools are used to check that control and treatment groups are the same on average?

- Checks and balance tables
 - performs a mean-difference test between treatment and control for
 - variables that are (should be) independent of treatment
- No significant differences (failing to reject H_0) suggest that the two groups are the same on average
- Any significant differences (rejecting H_0 at the α level of significance) imply that the two groups are not random.

Ceteris Paribus

Other things are equal! The “other things” are:

- **measurable features** (age, height, costs, ...);
- **measurable but absent from our data**;
- **nonmeasurable factors** (ability, effort, connections, ...)

Checks and balances table

Source: (Miguel and Kremer 2004)
table 1.

Variable-specific means are not the same across groups.

- How to conclude about any difference?

Note: *** indicates p-value < 0.01, ** p-value < 0.05, and * p-value < 0.1.

	Group 1 (25 schools)	Group 2 (25 schools)	Group 3 (25 schools)	Group 1 – Group 3	Group 2 – Group 3
<i>Panel A: Pre-school to Grade 8</i>					
Male	0.53	0.51	0.52	0.01 (0.02)	–0.01 (0.02)
Proportion girls <13 years, and all boys	0.89	0.89	0.88	0.00 (0.01)	0.01 (0.01)
Grade progression (= Grade – (Age – 6))	–2.1	–1.9	–2.1	–0.0 (0.1)	0.1 (0.1)
Year of birth	1986.2	1986.5	1985.8	0.4** (0.2)	0.8*** (0.2)
<i>Panel B: Grades 3 to 8</i>					
Attendance recorded in school registers (during the four weeks prior to the pupil survey)	0.973	0.963	0.969	0.003 (0.004)	–0.006 (0.004)
Access to latrine at home	0.82	0.81	0.82	0.00 (0.03)	–0.01 (0.03)
Have livestock (cows, goats, pigs, sheep) at home	0.66	0.67	0.66	–0.00 (0.03)	0.01 (0.03)
Weight-for-age Z-score (low scores denote undernutrition)	–1.39	–1.40	–1.44	0.05 (0.05)	0.04 (0.05)
Blood in stool (self-reported)	0.26	0.22	0.19	0.07** (0.03)	0.03 (0.03)
Sick often (self-reported)	0.10	0.10	0.08	0.02** (0.01)	0.02** (0.01)
Malaria/fever in past week (self-reported)	0.37	0.38	0.40	–0.03 (0.03)	–0.02 (0.03)

Hypothesis testing

Formulate H_0 and H_1 .

- $H_0 : \text{diff} = 0$
- $H_1 : \text{diff} \neq 0$

Use a statistical test, called **t-test**

- to conclude whether to reject H_0 or fail to reject H_0 .

t-tests (1)

Is a statistical test use to conclude about the population based on a sample.

1. One sample t-test: Compares a population parameter μ to an arbitrary number?
2. Two sample t-test: Compares difference between two parameters μ_1 and μ_2 in the population?

Its purpose is to infer:

- whether or not the population parameter μ (mean, or else)
- is different from an arbitrary number (usually zero: $\mu_0 = 0$)
- based on our sample estimates of
 - the parameter (mean), its variance (s), and the sample size n .

t-tests (2) Algorithm 1/2

Step 1: Formulate our **Hypotheses**

$$H_0: \mu = 0$$

$$H_1: \mu \neq 0$$

Step 2: Compute out **Statistics** based on the sample

- Estimated parameter: $\hat{\mu}$
- Standard Deviation: s
- Zero hypothesis value: $\mu_0 = 0$
- Significance level: $\alpha = 0.05$
- Number of observations: n

t-tests (3) Algorithm 2/2

Step 3: Compute **t-statistics** (and a **p-value** using the PC)

$$t = \frac{\hat{\mu} - \mu_0}{se} = \frac{\sqrt{n}(\hat{\mu} - \mu_0)}{s},$$

where $se = s/\sqrt{n}$ is the standard error of the estimate;

Step 4: **Conclude**

1. Reject H_0 if:

- $|t| > 2$ the “rule of thumb”
 - The same applies if we say that the estimate is two times larger than the mean.
- $p\text{-value} < \alpha$ (probability of type I error < level of significance)

2. Fail to reject H_0

Checks and balances: conclusions

Are the treatment and control groups on average similar?

Table source: ([Miguel and Kremer 2004](#)).

Note: *** indicates p-value < 0.01, ** p-value < 0.05, and * p-value < 0.1.

If yes: A mean comparison of the outcome variables reveals the causal effect

If no: We need to use other methods to measure the effect, such as regression.

TABLE I
1998 AVERAGE PUPIL AND SCHOOL CHARACTERISTICS, PRE-TREATMENT^a

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t-tests (3.1) example

Do the CCTs improve children school enrollment? ([Fiszbein et al. 2009](#))

^

^

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- The effect: Enrollment change = After the CCT – Before the CCT
- Hypothesis:
 - H_0 : Enrollment change = 0
 - H_1 : Enrollment change > 0
- Statistical test:
 - One-sided t-test
 - at the 5% level of significance $\alpha = 0.05$
- **Desired t-test result:**
 - Which t-test result indicates that CCT has a positive effect on school enrollment?
 - p-value < α or p-value < 0.05
 - $|t\text{-statistics}| > c_{df, 0.95}$
 - We need to be able to reject H_0

t-tests (3.2) example

Table 5.1 Impact of CCTs on School Enrollment and Attendance, Various Years

Country	Program	Age/Gender/Grade	Baseline enrollment (%)	Impact ^a	Transfer (% of PCE) ^b	Evaluation method	Reference
<i>Latin American and Caribbean countries</i>							
Chile	Chile Solidario	Ages 6–15	60.7	7.5*** (3.0)	7	RDD	Galasso (2006)
Colombia	Familias en Acción	Ages 8–13	91.7	2.1** (1.0)	17	PSM, DD	Attanasio, Fitzsimmons, and Gómez (2005)
		Ages 14–17	63.2	5.6*** (1.8)			
Ecuador	Bono de Desarrollo Humano	Ages 6–17	75.2	10.3** (4.8)	10	IV, randomized	Schady and Araujo (2008)
Honduras	Programa de Asignación Familiar	Ages 6–13	66.4	3.3*** (0.3)	9	Randomized	Glewwe and Olinto (2004)
Jamaica	Program of Advancement through Health and Education	Ages 7–17	18 days ^c	0.5** (0.2)	10	RDD	Levy and Ohls (2007)
Mexico	Oportunidades	Grades 0–5	94.0	1.9 (25.0)	20	Randomized	Schultz (2004)
		Grade 6	45.0	8.7*** (0.4)			
		Grades 7–9	42.5	0.6 (56.4)			
Nicaragua	Atención a Crisis	Ages 7–15	90.5	6.6*** (0.9)	18	Randomized	Macours and Vakis (2008)
Nicaragua	Red de Protección Social	Ages 7–13	72.0	12.8*** (4.3)	27	Randomized	Maluccio and Flores (2005)

- The column for “impact” reports the difference and standard error (in parentheses); the unit is percentage points, except of the Jamaican PATH program.
- *, **, and *** indicates if the p-value is below a corresponding significance levels of 10, 5, and 1 percent.
- What is shown in the rows?
 - case-specific t-test
- What are H_0 and H_1 ?
 - $H_0: \text{diff} = 0; H_1: \text{diff} \neq 0;$
- What are the conclusions for each row?
 - Reject/Accept H_0
- What do “stars” imply for test-specific conclusions for each row?

t-tests (3.3) example

Table 5.1 continued

Country	Program	Age/Gender/ Grade	Baseline enrollment (%)	Impact ^a	Transfer (% of PCE) ^b	Evaluation method	Reference
<i>Non-Latin American and Caribbean countries</i>							
Bangladesh	Female Secondary School Assistance Program	Ages 11–18 (girls)	44.1	12.0** (5.1)	0.6	FE	Khandker, Pitt, and Fuwa (2003)
Cambodia	Japan Fund for Poverty Reduction	Grades 7–9 (girls)	65.0	31.3*** (2.3)	2–3	DD	Filmer and Schady (2008)
Cambodia	Cambodia Education Sector Support Project	Grades 7–9	65.0	21.4*** (4.0)	2–3	RDD	Filmer and Schady (2009c)
Pakistan	Punjab Education Sector Reform Program	Ages 10–14 (girls)	29.0	11.1*** (3.8)	3	DDD	Chaudhury and Parajuli (2008)
Turkey	Social Risk Mitigation Project	Primary school	87.9	–3.0* n.a.	6	RDD	Ahmed et al. (2007)
		Secondary school	39.2	5.2 n.a.			

Source: Authors' compilation.

Note: DD = difference-in-differences; DDD = difference-in-difference-in-differences; FE = fixed effects; IV = instrumental variables; n.a. = not available; PCE = per capita expenditure; PSM = propensity score matching; RDD = regression discontinuity design. This table contains unweighted means for the coefficients for Colombia ages 8–13 and 14–17, Chile ages 4–5 and 6–15, and Mexico grades 0–5 and 7–9. The standard errors in each case are the square roots of the averaged variances of these estimates.

a. The column for “impact” reports the coefficient and standard error (in parentheses); the unit is percentage points, with the exception of the Jamaican PATH program, where the unit is days.

b. The transfer amounts as a proportion of per capita expenditures (or consumption) are not the same across all tables in the report because of differences in the surveys used, including their coverage and year.

c. Impacts were measured in Jamaica only for student attendance over a 20-day reference period. The baseline enrollment rate prior to PATH was 96 percent.

Interpret results of the statistical test for country A at the X (1%, 5%) level of significance.

- State Null and Alternative hypotheses.
- Conclude about the HT results.
- What parameter/statistics/number was used to make this conclusion?

For example. In (Ahmed et al. 2007) authors failed to show that CCTs improve/worsen children school enrollment in Turkey. With “H0: no improvement” versus the “H1: some improvement”, the authors failed to reject the H0 at the 5% sign. level for the populations of primary and secondary school attendees ...

(For example cont.) ... In absolute terms, authors show a reduction in primary school enrollment by 3% and an increase in secondary school enrollment by 5.2%. However, the probability that these changes are due to sample variation (probability of type I error / p-value) is above 5% and 10% respectively. Table source: (Fiszbein et al. 2009)

Homework: Read/do more on Hypothesis testing

[OpenIntro](#) Foundations for Inference:

- [Point Estimates](#)
- [Hypothesis Testing Fundamentals](#)
- [Inference for Estimators Other Than the Mean](#)

Khan Academy:

- [The idea of significance tests](#)

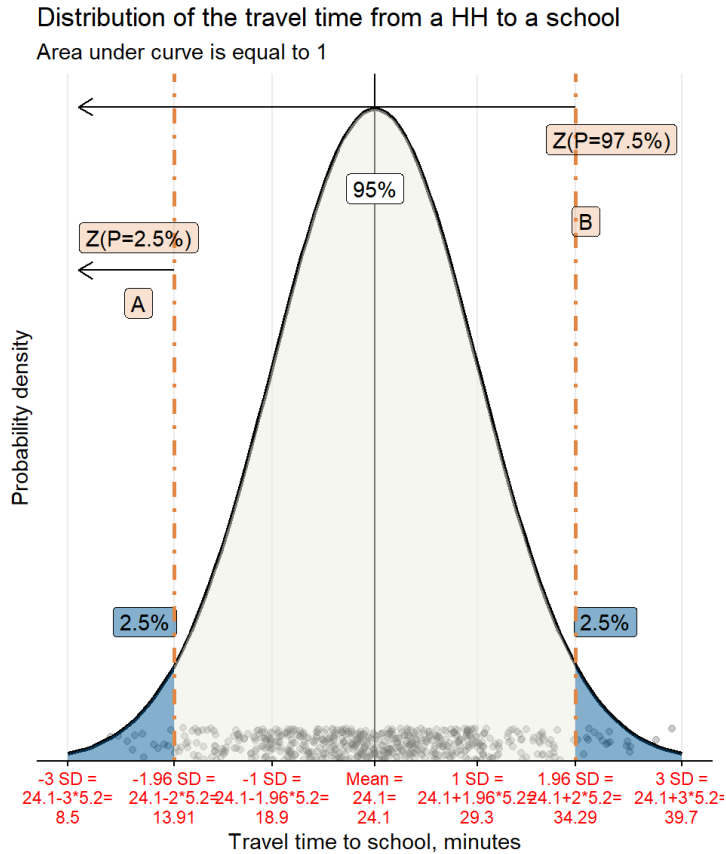
Normally distributed population and the confidence intervals

Suppose that we have measured travel time to school for the entire population of the HHs.

Travel time is distributed normally with:

- μ - average travel time;
- σ - standard deviation of μ ;

Population distribution



Travel time distribution is normal with mean μ and SD σ :

- Within 2σ , there are 95.45% of all observations.

Within what number of σ , exactly 95% of all observations are? Any ideas?

95% of HH are located within $\pm 1.96\sigma$.

- 1.96 is the z-score, where:

A. probability to observe travel time below $\mu - 1.96\sigma$

- $P(x < \mu - 1.96\sigma) = 0.025$

B. probability to pick a HH with a travel time below $\mu + 1.96\sigma$

- $P(x < \mu + 1.96\sigma) = 0.975$

95% Confidence interval for the mean is

- an interval (a range), within which we can observe 95% of observations.
- 95% is the **confidence level**
- 95% confidence level is a convention. Go higher, not lower!

Warning

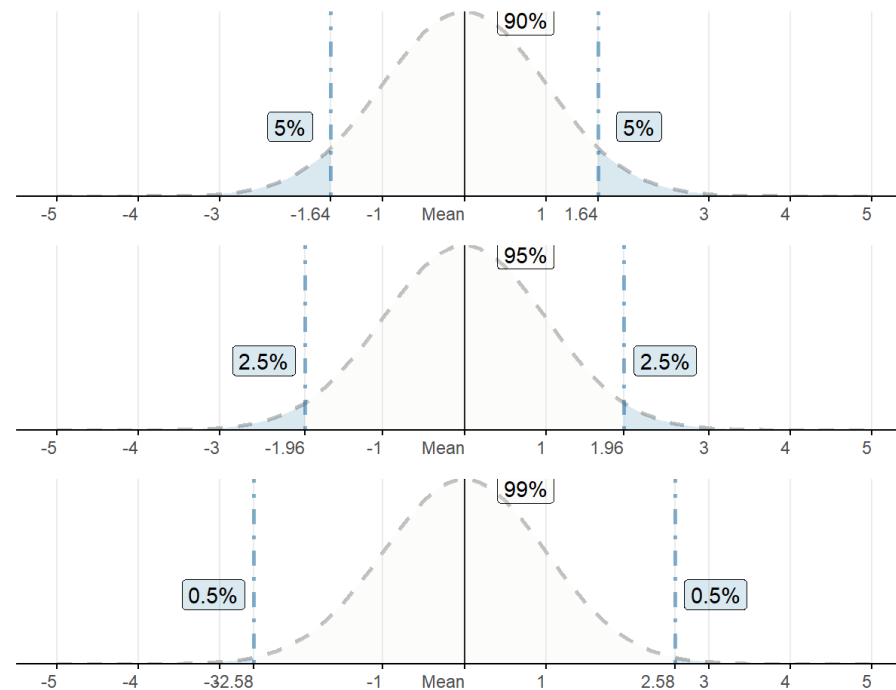
This definition only applies to the **population mean**.

Although the idea behind a confidence interval remains the same.

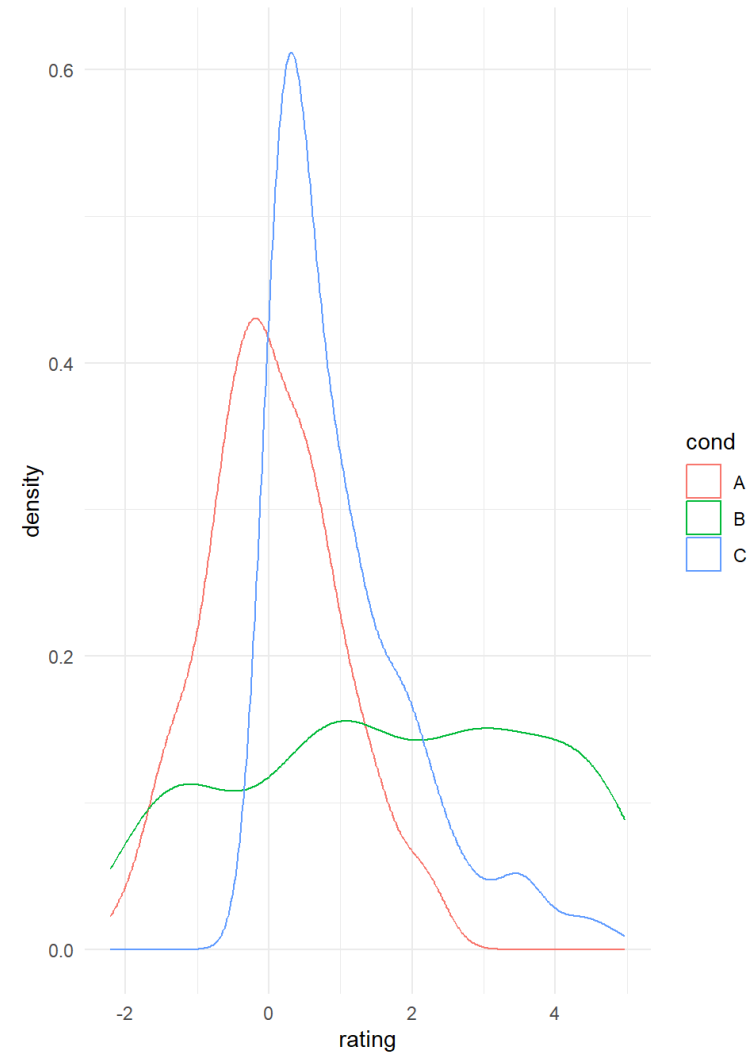
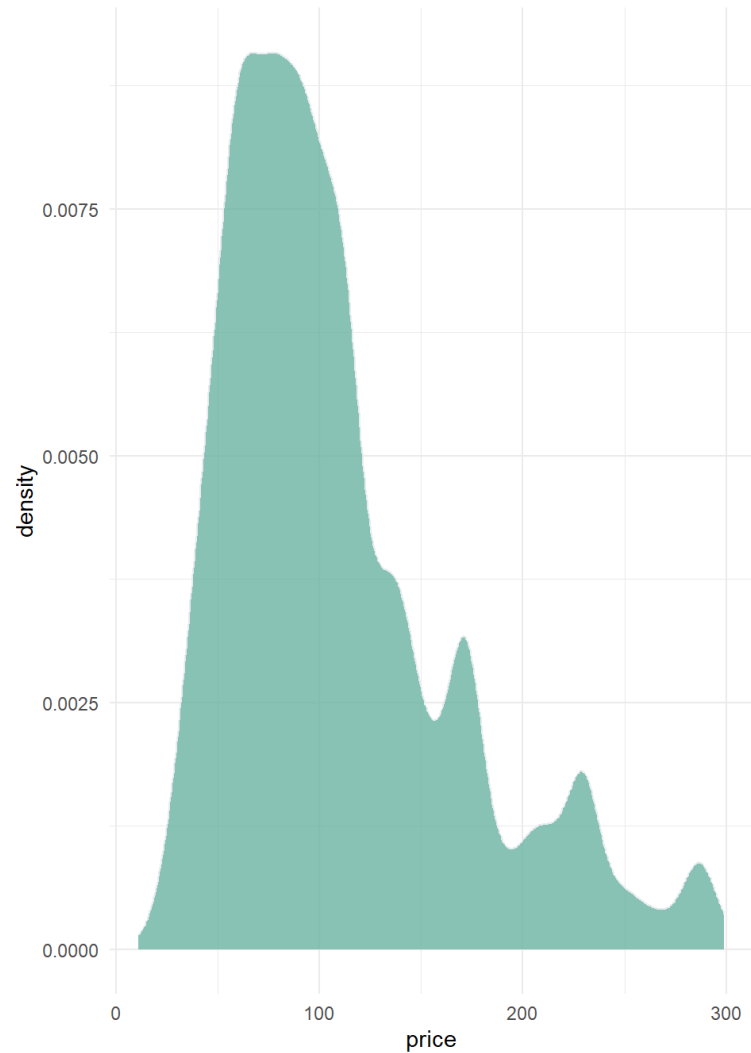
90%, 95% and 99%, Confidence intervals in the population

Is 90% confidence interval wider/slimmer than the 95% one?

99% vs 95%?



The population can have any distribution



The Central Limit Theorem (CLT)

The CLT is the bridge that makes the transition from the population to a sample.

As the sample is used to **estimate** the population mean, the CLT says that:

- **almost any sample average**
- is approximately normally distributed,
- with the accuracy of the approximation increasing as the sample size increases ([Angrist and Pischke 2014, 39](#))

The Central Limit Theorem (2)

The estimate of the mean $\hat{\mu}$ in the sample of the size n

- tends to follow a normal distribution centered at the population's mean μ ,
- with the standard error, $SE(\hat{\mu})$ calculated based on the population's standard deviation, $SE(\hat{\mu}) = \sigma / \sqrt{n}$,
- when two conditions are satisfied:
 1. **Independence.** The sample is random.
 2. Only for sample size below 30 obs ($n < 30$) **Normality.** The population is normally distributed.
- See: ([Diez, Cetinkaya-Rundel, and Barr 2022](#))

The Central Limit Theorem (3)

In other words:

- When our **random sample is large enough**
 - (or the population is normally distributed for small samples),
- **sample estimate of mean $\hat{\mu}$** approximates the population mean μ :
 - $\hat{\mu} \approx \mu$;
- with the **standard error** that depends on the σ (pop. SD) and the sample size (n):
 - $SE(\hat{\mu}) = \sigma / \sqrt{n}$.

The Central Limit Theorem (4)

Because the population's SD (σ) is unknown, we rely on the sample to estimate:

- the standard error:

- $SE(\hat{\mu}) = \sigma / \sqrt{n} \approx \overset{\wedge}{SE}(\hat{\mu}) = s / \sqrt{n},$

- where $\sigma \approx s;$

- alone with the estimate of mean: $\hat{\mu} \approx \mu;$

Standard Error (1)

SE is an **estimate** of the standard deviation **in the sampling distribution** of $\hat{\mu}$.

$$\text{SE}(\hat{\mu}) = s / \sqrt{n}$$

Measures deviation (error) **of the** $\hat{\mu}$ estimate, from the population parameter (μ), accounting for the size of the sample (n).

Standard Error (2)

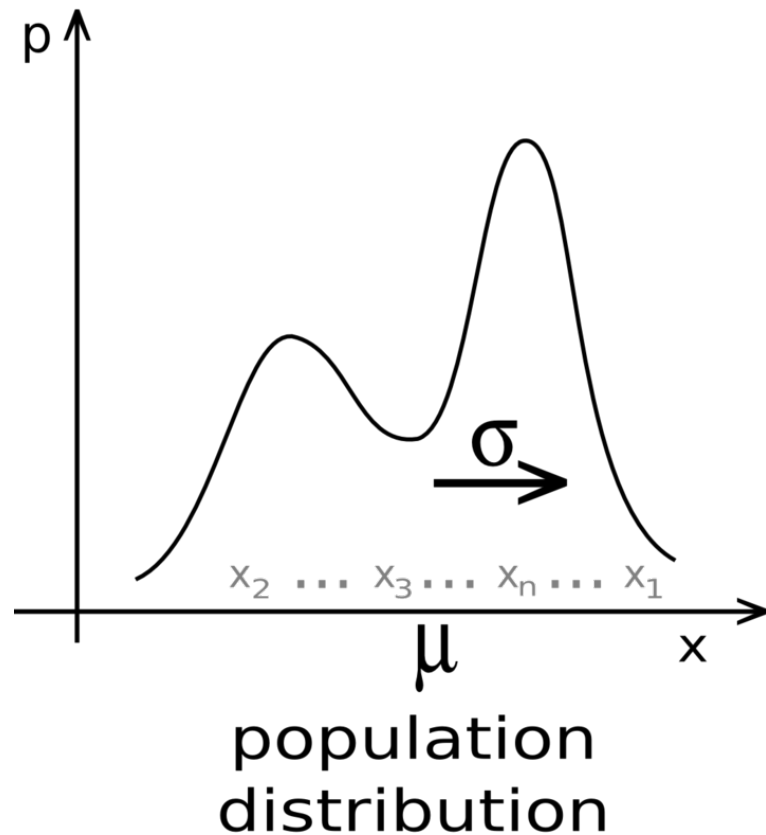
“Error” implies that **the estimate** contains a **random error**, compared to the population parameter.

- Not to be confused with the **Sample Variance**

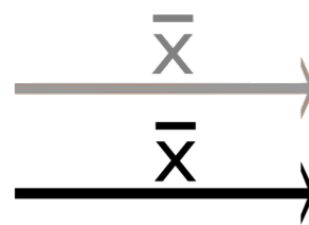
- $s^2 = \frac{\sum_{i=1}^N [x_i - \bar{x}]^2}{N-1}$

- With the sample size increasing to infinity:
 - **standard error approaches to zero,**
 - **but the sample variance does not!**

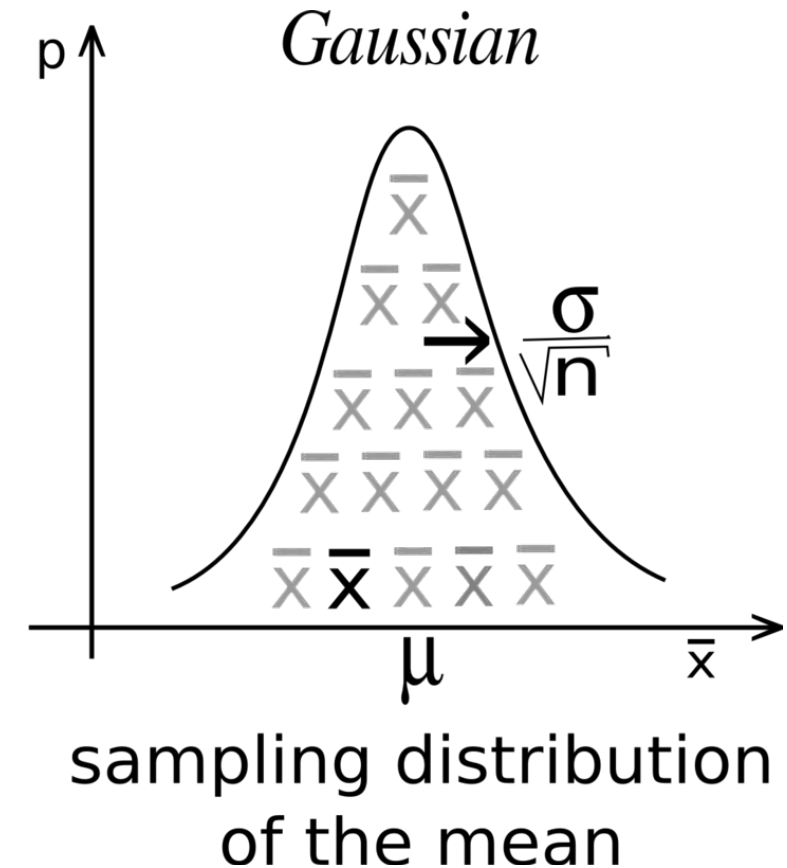
CLT summary (from <en.wikipedia.org/wiki/Central_limit_theorem>)



samples
of size n



Two horizontal arrows pointing to the right. The top arrow is light gray and labeled $\bar{x}$. The bottom arrow is black and labeled $\bar{x}$.



CLT in practice

See the following online tools:

- [CLT in practics](#)

The Confidence Interval

Supposed that we have a population of unknown distribution, and

- we derive a random sample of a size n and
- estimate sample mean $\hat{\mu}$ and $SE(\hat{\mu})$

The Confidence Interval is the range around the sample mean, within which there is a 95% (or 99%) chance of observing our population mean.

How to find a CI?

We need to know:

- how the sample means are distributed,
- when we repeat random sampling from the same population.

The CLT [is very instrumental because it] says:

- sample means are distributed normally, and
- are centered around the population mean,
- when the sample size is large (infinite)

But random samples are finite and often small in size...

The t-distribution

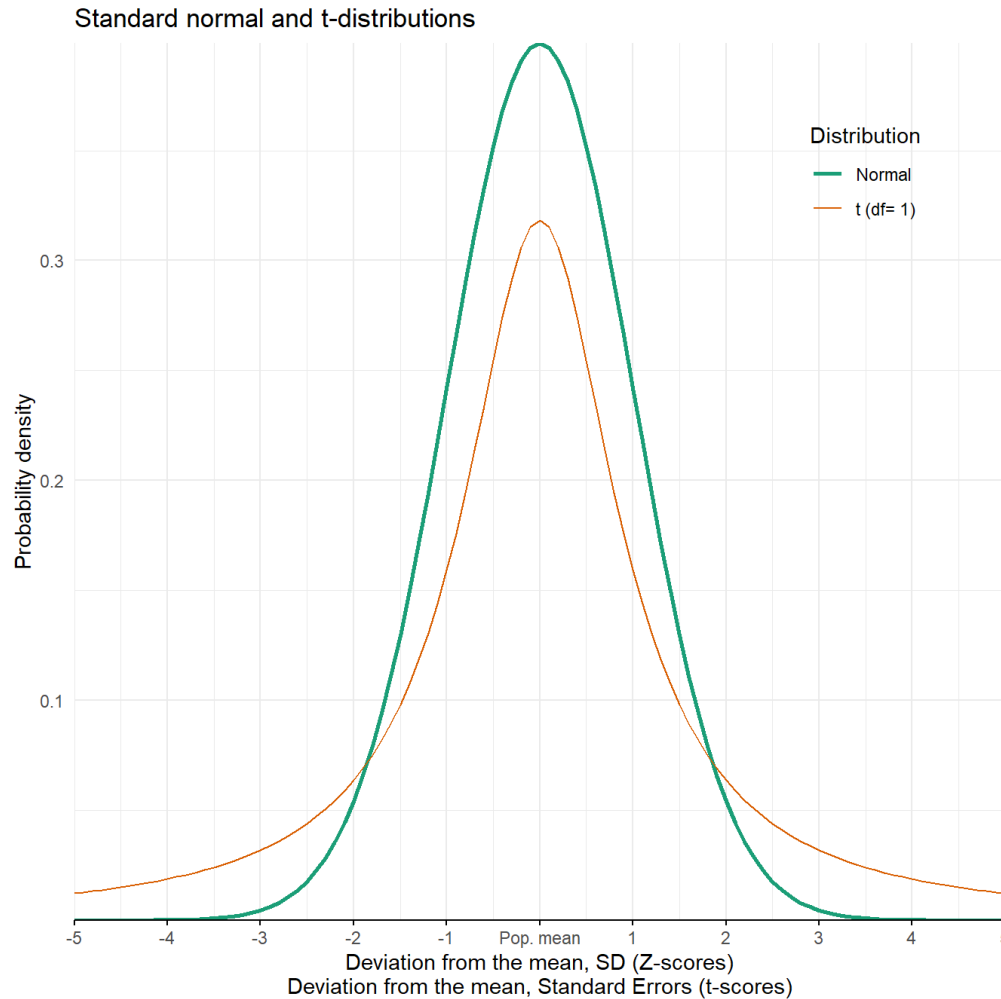
We use **t-distribution** (instead of the normal) to adjust for the finite sample size and

- to describe the probability distribution of sample mean ($\hat{\mu}$) with variance ($SE(\hat{\mu})$).

t-distribution helps us:

- to find the probability that the true population mean is beyond $\pm SE(\hat{\mu})$ from the $\hat{\mu}$, conversely
- tells us how many $\pm SE(\hat{\mu})$ from the $\hat{\mu}$ there is a 95% chance to observe the population mean for the sample of the size n .

From the population to a sample (1)

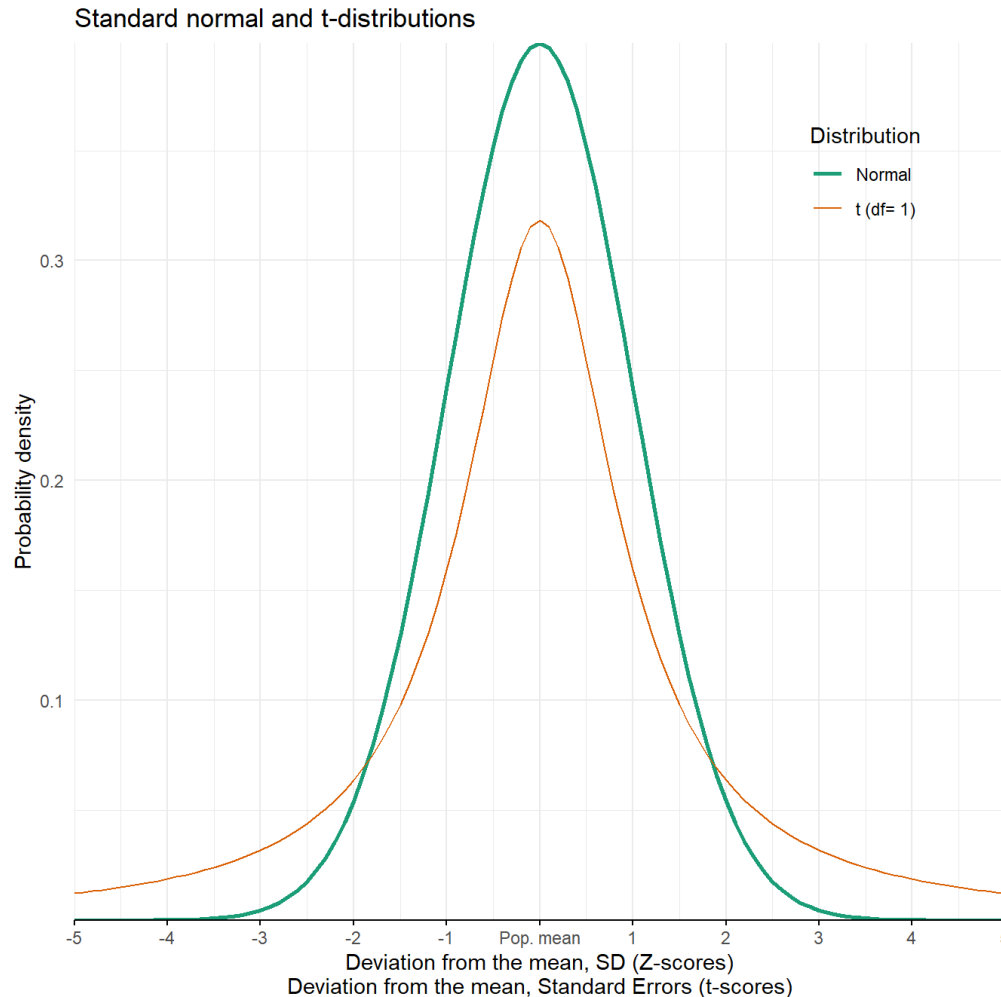


Sample **estimates** are **less accurate** when the size is small.

Therefore, for inference,

- we use **t-distribution** instead of the normal.
- t-scores instead of the z-scores

t-distribution (1)



t-distribution has “thicker tails”:

- the probability of observing a more extreme parameter, compared to the population value is larger;
- the peak of t-distribution does not go too high;
- observations are more likely to fall beyond 2SE away from the mean;

t-distribution (2)

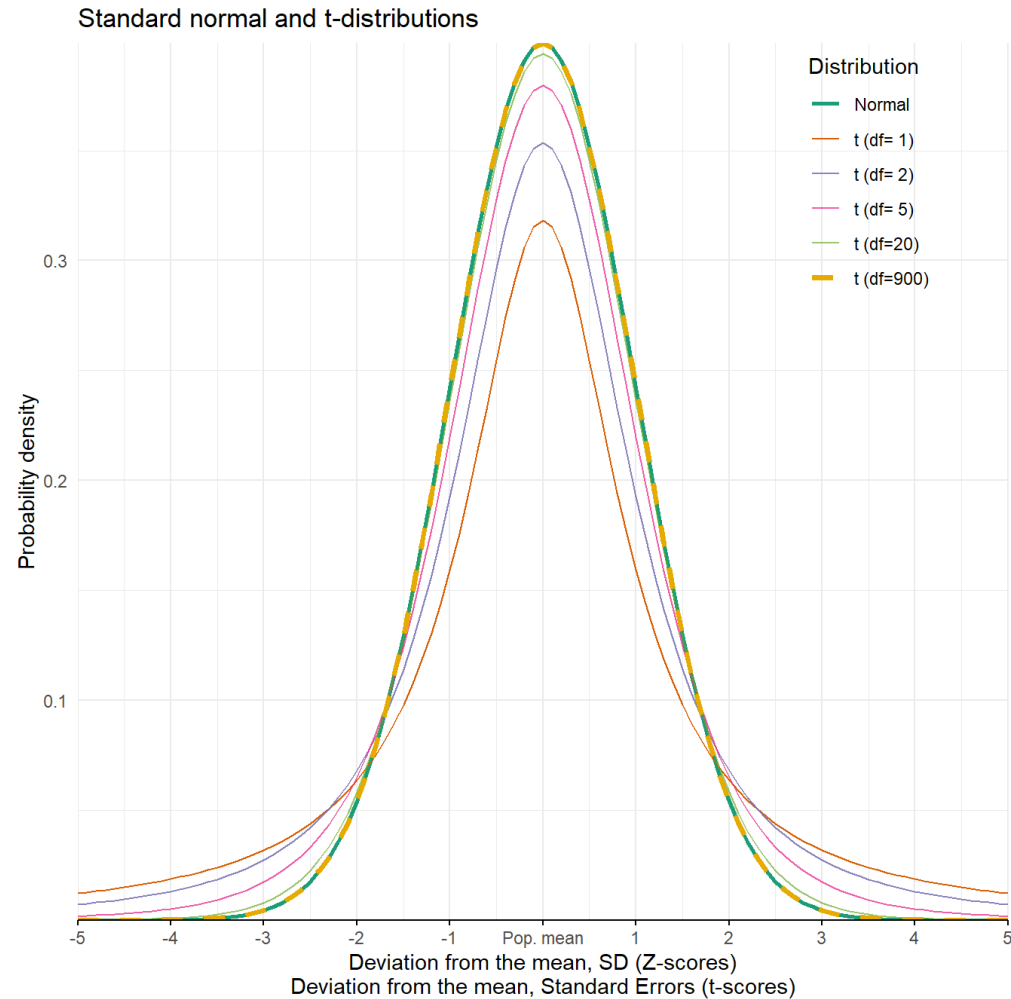
The shape of the t-distribution depends on **degrees of freedom**:

$df = n - k$, where

n number of observations in the sample; k number of estimated parameters;

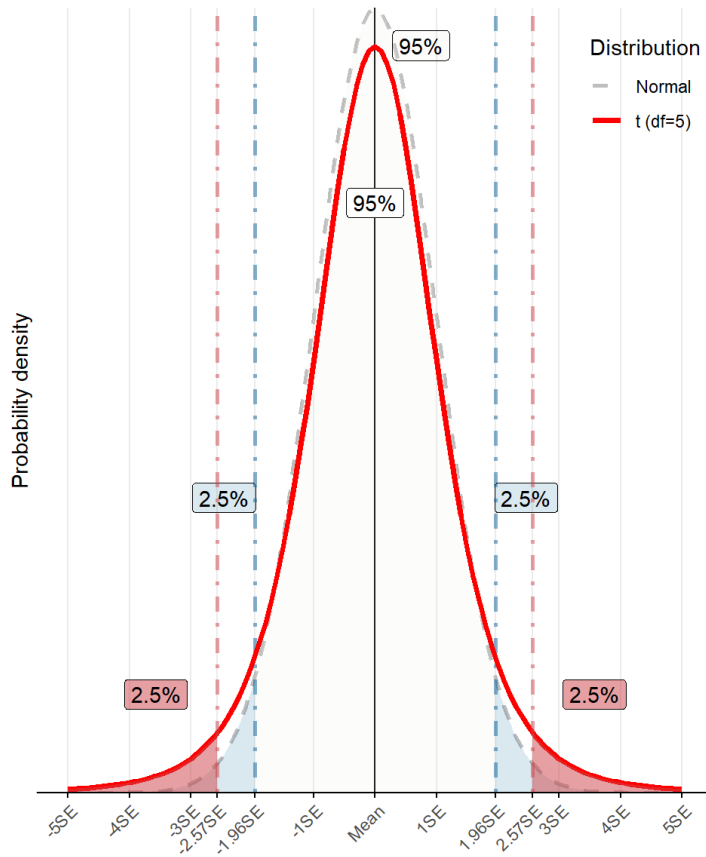
With an increase in df ,

- t-distribution becomes similar to the normal distribution.

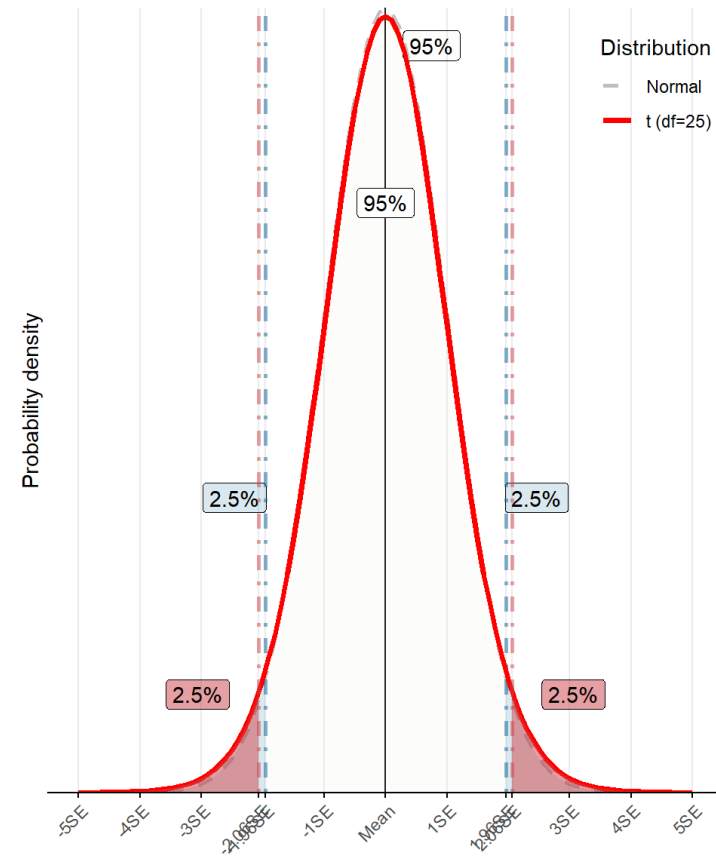


t-distribution (3)

Confidence intervals for different distributions
Each area under curve is 1

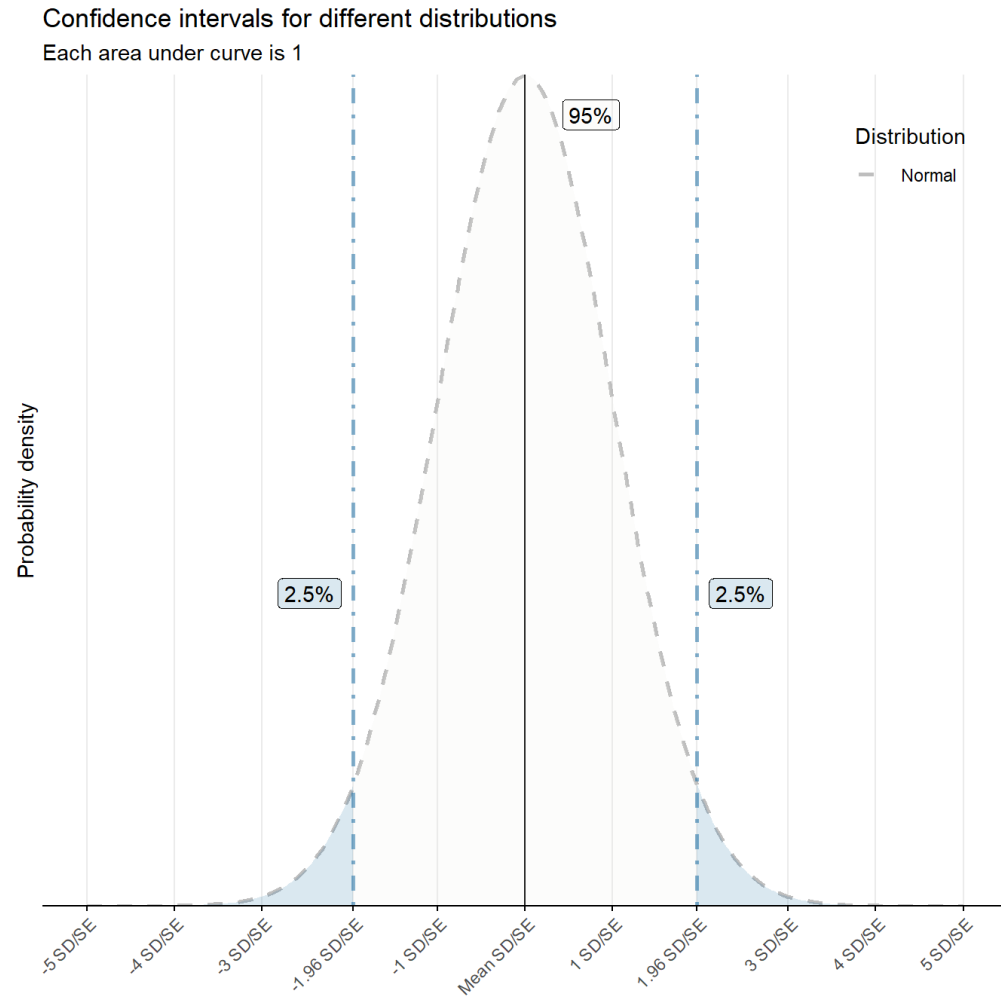


Confidence intervals for different distributions
Each area under curve is 1



Application of the t-distribution

Recall the travel time example



For the population:

- the standard normal distribution and Z-score provide
- the **confidence intervals**, where we can observe 95% of actual observations.

t-dist in use: travel time (1)

We have a population of travel times to schools.

Let us now:

1 Derive a random sample of N observations.

2 Estimate $\hat{\mu}$ and $SE(\hat{\mu})$

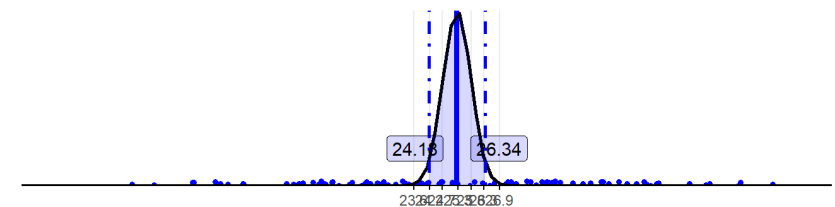
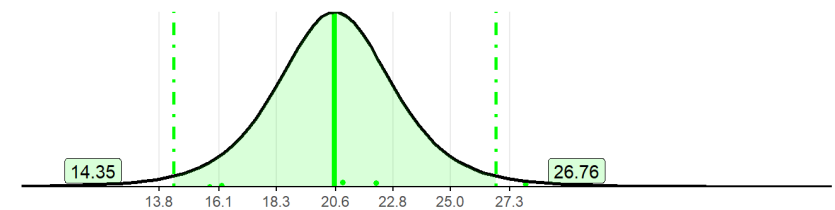
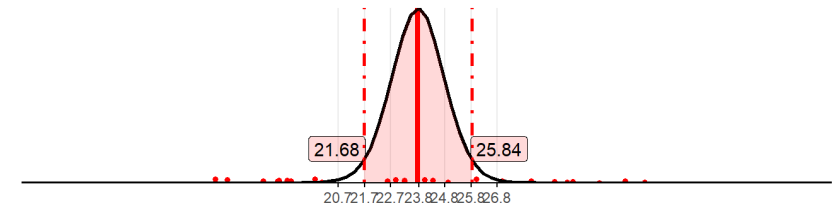
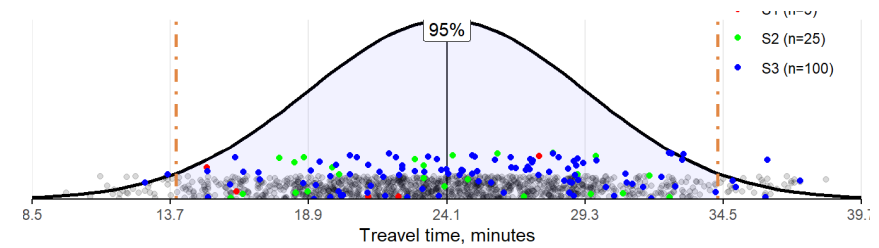
Sample	n	$\hat{\mu}$	s	SE
S1	25	23.76	5.04	1.01

3 Construct the t-distribution for the sample estimates of the mean.

4 Compute **95% confidence intervals**: chances that the population mean is within this range are 95%;

5 Conclude that 95% CI is the range where the population mean is 95% likely to be given our sample size, $\hat{\mu}$ and $SE(\hat{\mu})$

Now, let us repeat steps 1-5 for new samples of different sizes.

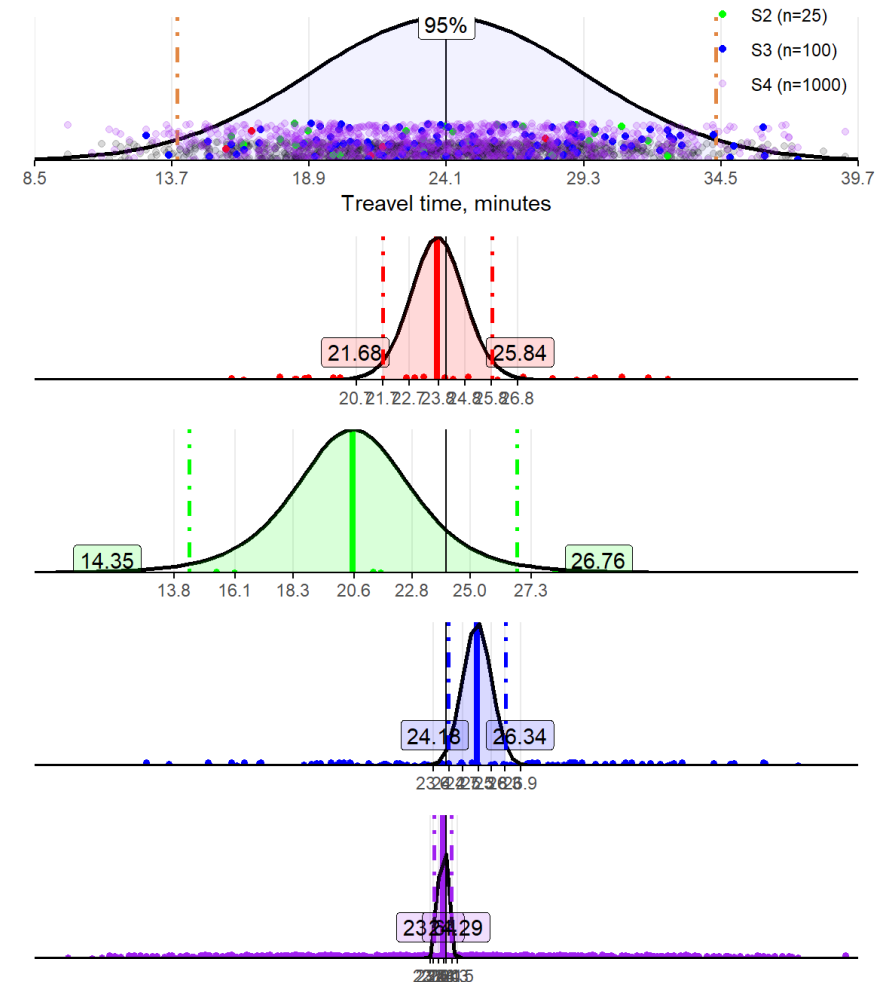


t-dist in use: confidence intervals

There are four samples of different size:

Each contains estimated parameters, which are different from the population.

Sample	n	$\hat{\mu}$	s	SE
S1	25	23.76	5.04	1.01
S2	5	20.55	5.00	2.24
S3	100	25.26	5.43	0.54
S4	1000	23.97	5.25	0.17
Pop.	Inf	24.10	5.20	0.00



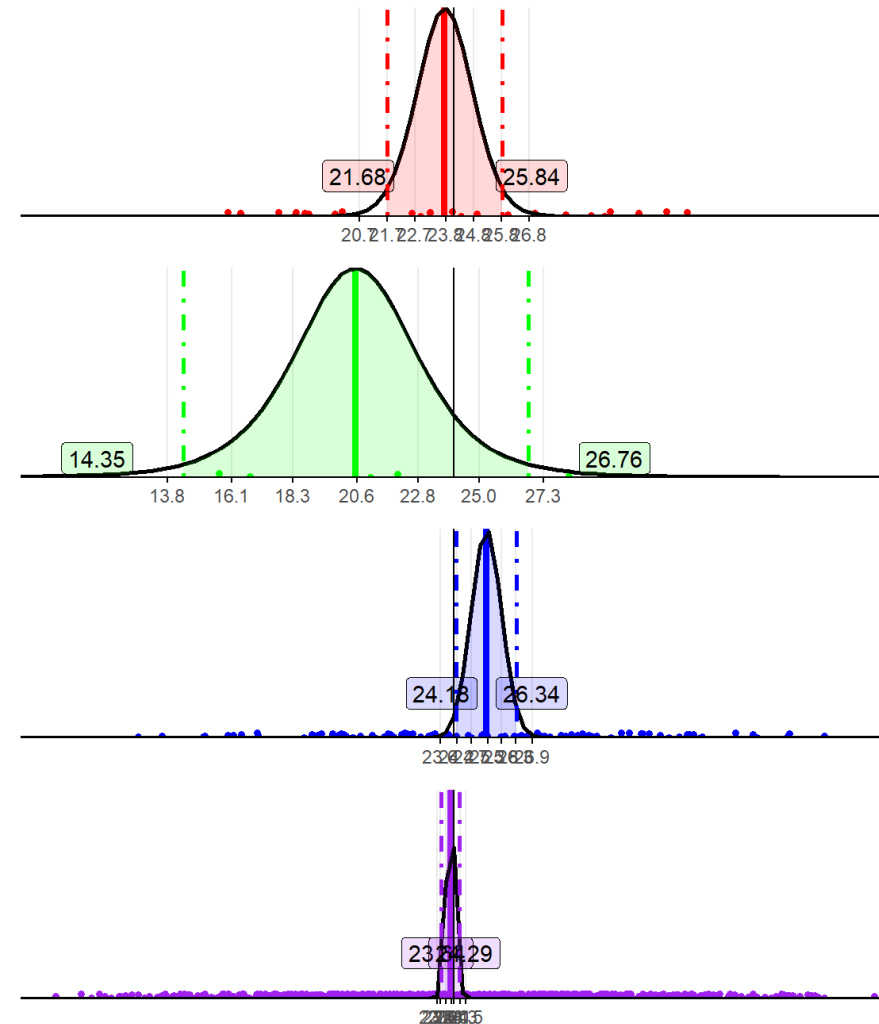
⚠ Using t-distribution, for each sample of size n we compute a **95% confidence interval**.

A **95% confidence interval** provides a range, within which the population mean lies with the probability of 95%.

95% Confidence intervals of the mean

95% confidence interval of the mean is:

- the range, within which we may find the population mean in 95% of random samples made from the same population.
- we are 95% confident in observing the population parameter given the current sample size and df



90%, 95%, and 99% Confidence intervals of the mean

90%, 95%, and 99% are different **levels of confidence**;

- Confidence level (CL) is the opposite of the level of significance α :
 - $CL = 1 - \alpha$
- It is possible to compute confidence intervals at a different CL.
- They will change in the same way as CI for the normally distributed variable shown before.

Interpreting Confidence Interval

Source: <https://rpsychologist.com/d3/ci/>

Calculating a Confidence Interval (1)

$P[\dots]$ - the probability that the population mean μ lies within the **lower and upper** margins of the confidence interval:

$$P[\hat{\mu} - c_{df, (1-\alpha/2)} \cdot SE(\hat{\mu}) \leq \mu \leq \hat{\mu} + c_{df, (1-\alpha/2)} \cdot SE(\hat{\mu})] = 1 - \alpha$$

- $\hat{\mu}$ - estimated mean;
- $SE(\hat{\mu})$ - the standard errors;
- α - level of significance (as before);
- μ - unknown population mean;
- $c_{df, 1-\alpha/2}$ - the critical value of the t distribution given α , df and two-tailed test nature;
- df - number of degrees of freedom (as before);

$$CI = \{ \hat{\mu} - c_{df, (1-\alpha/2)} \cdot SE(\hat{\mu}) ; \hat{\mu} + c_{df, (1-\alpha/2)} \cdot SE(\hat{\mu}) \}$$

Calculating a Confidence Interval (2)

Sample of the size 5 is derived from the population. Estimated statistics for the travel time from HHs to school in minutes are:

Sample	<i>n</i>	$\hat{\mu}$	<i>s</i>	SE
S2	5	20.55	5	2.24

Calculate the 95 confidence intervals for this mean.

df is: 4

$c_{df, (1-\alpha/2)}$ is: $c_{4, (1-0.05/2)} = 2.776$

Lower bound is: $20.55 - 2.24 * 2.776 = 14.331$

Upper bound is: $20.55 + 2.24 * 2.776 = 26.769$

CI is: [14.331 ; 26.769]

df	0.9	0.95	0.975
1	3.0777	6.3138	12.7062
2	1.8856	2.9200	4.3027
3	1.6377	2.3534	3.1824
4	1.5332	2.1318	2.7764
5	1.4759	2.0150	2.5706
6	1.4398	1.9432	2.4469
7	1.4149	1.8946	2.3646
8	1.3968	1.8595	2.3060
9	1.3830	1.8331	2.2622
15	1.3406	1.7531	2.1314
20	1.3253	1.7247	2.0860
25	1.3163	1.7081	2.0595
50	1.2987	1.6759	2.0086
100	1.2901	1.6602	1.9840
1000	1.2824	1.6464	1.9623

Calculating a Confidence Interval (3)

Sample of the size 25 is derived from the population. Variable measured travel time from HHs to school in minutes. Estimated statistics are:

Sample	n	$\hat{\mu}$	s	SE
S1	25	23.76	5.04	1.01

Calculate the 99 confidence intervals for this mean.

df is: 4 $c_{4, (1-0.01/2)}$ is: 2.064

Lower bound is: $23.76 - 1.01 * 2.064 = 21.675$

Upper bound is: $23.76 + 1.01 * 2.064 = 25.845$

CI is: 21.675 ; 25.845

df	0.975	0.99	0.995
1	12.7062	31.8205	63.6567
2	4.3027	6.9646	9.9248
3	3.1824	4.5407	5.8409
4	2.7764	3.7469	4.6041
5	2.5706	3.3649	4.0321
6	2.4469	3.1427	3.7074
7	2.3646	2.9980	3.4995
8	2.3060	2.8965	3.3554
9	2.2622	2.8214	3.2498
15	2.1314	2.6025	2.9467
20	2.0860	2.5280	2.8453
25	2.0595	2.4851	2.7874
50	2.0086	2.4033	2.6778
100	1.9840	2.3642	2.6259
1000	1.9623	2.3301	2.5808

More on the t-distribution and t-test

Open intro:

7.1A - t-distribution

7.1B - Inference for one mean

7.2 - Paired data

7.3 - Difference of two means

Khan academy:

Module: Tests about a population mean

More significance testing videos

Practice

Example 1: Effect of a drug on sleep

(1)

Question: Does the drug increase sleeping time at the 5% level of significance?

- Null **hypothesis**: the drug has a zero effect
- Alternative: the drug has a nonzero effect

Data:

- sleep time for each patient before they started taking drugs (control)
- the same after they took drugs (treatment)
- for each patient separately, we computed a difference in sleep time between control and treatment.
- measures are made for 10 patients

The estimated parameters are:

vars	n	mean	sd	se
1	10	0.75	1.78901	0.5657345

- Can we conclude that the mean is different from zero?

Example 1: Effect of a drug on sleep (2)

Can we conclude that the mean is different from zero?

- **NO!** we cannot.

Because:

- the mean difference could **result from a small sample or sample's variation.**
- to conclude about the population, we need to take into account the sample distribution.

We need to perform a Statistical Hypothesis Testing with:

- The one-sample two-sided t-test.
- Compare mean to a zero;

One-sample t-test (1): What?

One-sample t-test

- H_0 mean is equal to the parameter μ specified (or zero if not specified);
- H_1 mean is not equal to zero;
- To decide whether to reject or accept the null hypothesis, we need to rely on:

```
1 t.test(slp$extra, mu = 0)
2 # One Sample t-test
3 # data: slp$extra
4 # t = 1.3257, df = 9, p-value = 0.2176
5 # alternative hypothesis: true mean is not equal to 0
6 # 95 percent confidence interval:
7 # -0.5297804 2.0297804
8 # sample estimates:
9 # mean of x
10 # 0.75
```

- The test of significance
- The confidence interval
- The p-value

One-sample t-test (2): The Test of Significance

1. Check the t statistics against the critical value $c_{df, 1-\alpha/2}$.

```
1 t.test(slp$extra, mu = 0)
2 # One Sample t-test
3 # data: slp$extra
4 # t = 1.3257, df = 9, p-value = 0.2176
5 # alternative hypothesis: true mean is not equal to 0
6 # 95 percent confidence interval:
7 # -0.5297804 2.0297804
8 # sample estimates:
9 # mean of x
10 # 0.75
```

- $df = 9$
- $c_{0.975, 9} = 2.2622$
- if $|t| \leq c_{df, 1-\alpha/2}$ - fail to reject H_0 ;
- if $|t| > c_{df, 1-\alpha/2}$ - reject H_0 in favor of H_1 at α significance level;

One-sample t-test (2): Confidence intervals

2. Check the confidence interval of the mean

```
1 t.test(slp$extra, mu = 0)
2 # One Sample t-test
3 # data: slp$extra
4 # t = 1.3257, df = 9, p-value = 0.2176
5 # alternative hypothesis: true mean is not equal to 0
6 # 95 percent confidence interval:
7 # -0.5297804 2.0297804
8 # sample estimates:
9 # mean of x
10 # 0.75
```

- if CI includes the value from H_0 (0) - fail to reject H_0 ;
- if CI does not include the number from H_0 , we reject H_0 in favor of H_1 at α significance level;

One-sample t-test (2): p-value

```
1 t.test(slp$extra, mu = 0)
2 # One Sample t-test
3 # data: slp$extra
4 # t = 1.3257, df = 9, p-value = 0.2176
5 # alternative hypothesis: true mean is not equal to 0
6 # 95 percent confidence interval:
7 # -0.5297804 2.0297804
8 # sample estimates:
9 # mean of x
10 # 0.75
```

- if $p - value \geq \alpha$, we fail to reject H_0 ;

- if $p - value < \alpha$, we reject H_0 in favor of H_1 at the $p - value$ significance level;

One-sample t-test (3): Conclusion

```
1 t.test(slp$extra, mu = 0)
2 # One Sample t-test
3 # data: slp$extra
4 # t = 1.3257, df = 9, p-value = 0.2176
5 # alternative hypothesis: true mean is not equal to 0
6 # 95 percent confidence interval:
7 # -0.5297804 2.0297804
8 # sample estimates:
9 # mean of x
10 # 0.75
```

- We fail to reject H_0
- The evidence is insufficient to conclude that the change in sleep time is significantly different from 0 at the 5% level of the sign.

Example 2: Travel time difference between treatment and control

We come back to the travel time example.

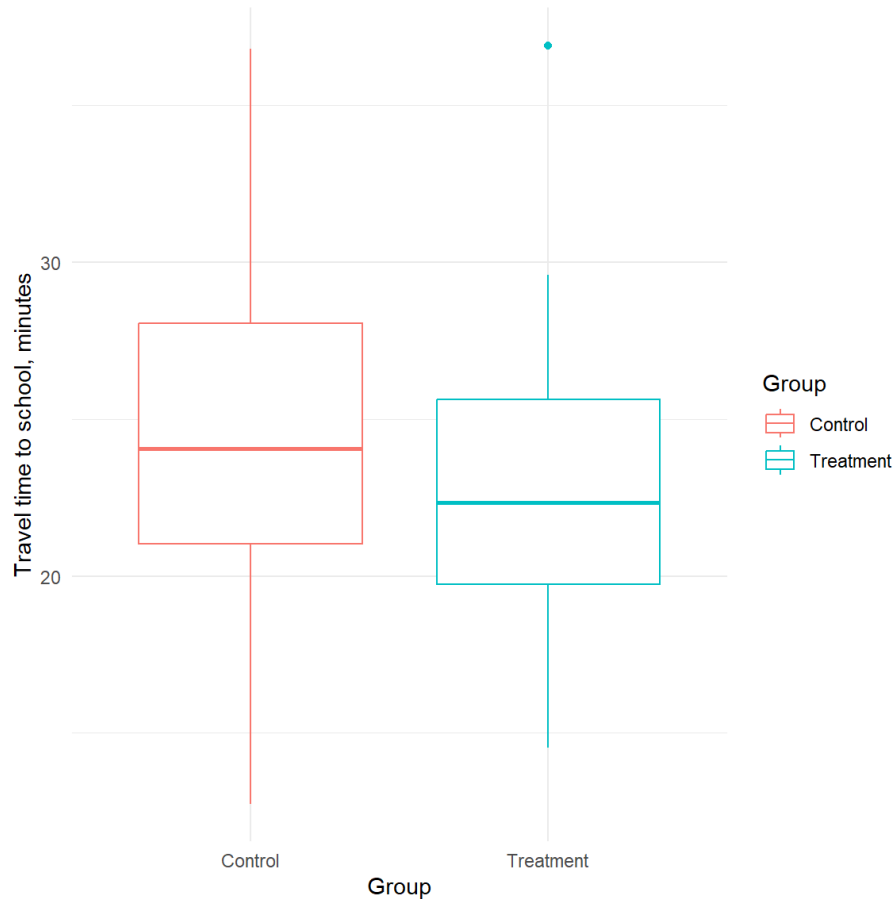
Checks and balances

Travel time example:

- Two random samples are derived from the population:
 - Treatment, where (CCTs) were provided;
 - Control, where no CCT was provided;

We want to establish that the mean travel times of these two groups are similar on average.

Exploratory analysis (1)



From the plot, we observe:

1. Medians are slightly different but they overlap within the boxes (for 50% of all observations)
2. Variances are different based on the whiskers. The presence of an outlier in the treatment may indicate that groups have similar variances, but the small sample size does not perfectly represent it.
3. We may consider using a t-test for two samples with unequal variances.

Exploratory analysis (2)

Group	n	$\hat{\mu}$	s	SE
Control	31	24.46	5.52	0.99
Treatment	33	23.01	4.64	0.81

Is there any difference between treatment and control?

- Absolute numbers are different.
- Given that these are two samples, we need to perform statistical Hypothesis Testing to establish if these two averages are statistically significantly different from one another.

Two-sample t-test (1)

```
1 t.test(`Travel time` ~ Group, data = pop_distance_sample,  
2       var.equal = FALSE)  
3 #    Welch Two Sample t-test  
4 #  
5 # data:  Travel time by Group  
6 # t = 1.1331, df = 58.767, p-value = 0.2618  
7 # alternative hypothesis: true difference in means between group Control and group Treatment is not eq  
8 # 95 percent confidence interval:  
9 #  -1.110563  4.009977  
10 # sample estimates:  
11 #    mean in group Control mean in group Treatment  
12 #           24.46403           23.01432
```

- Two-sample t-test with unequal variances between two samples.
- $H0: \hat{\mu}_1 = \hat{\mu}_2$ Two means are statistically similar (indifferent);
- $H1: \hat{\mu}_1 \neq \hat{\mu}_2$ Because the $H1$ is about an inequality sign, we have a two-tailed t-test ($c_{df, 1-\alpha/2}$)

Two-sample t-test (2) Conclusions

Critical value of the t-statistics (see full tables at the end of slides):

- $df = 58.767 \approx 58$
- $1 - \alpha/2 = 1 - 0.05/2 = 0.975$
- $c_{58,0.975} = 2.002$
- Line 6: $|t| < c_{58,0.975}$ as $|1.1331| < 2.002$
- Line 7: 95% confidence interval of difference between two means includes zero.

Conclusion:

- Fail to reject H_0 . The evidence is insufficient to conclude that average travel times between school and households in treatment and control groups are different at the 5% level of significance.

Two-sample t-test (3) Equal variance

```
1 t.test(`Travel time` ~ Group, data = pop_distance_sample,  
2       var.equal = TRUE)  
3 #    Welch Two Sample t-test  
4 #  
5 # data:  Travel time by Group  
6 # t = 1.1331, df = 58.767, p-value = 0.2618  
7 # alternative hypothesis: true difference in means between group Control and group Treatment is not eq  
8 # 95 percent confidence interval:  
9 #  -1.110563  4.009977  
10 # sample estimates:  
11 #    mean in group Control mean in group Treatment  
12 #          24.46403          23.01432
```

Test statistics, degrees of freedom, and confidence intervals change, but the conclusion remains the same.

Example 3: Causal effect of treatment on enrollment

The dependent variable here is a dummy variable (0|1), which takes the value of 1 if children are being enrolled in the school in a household and zero otherwise.

Descriptive statistics

Descriptive statistics of school enrollment for two groups.

Group	n	$\hat{\mu}$	s	SE
Control	31	0.290	0.461	0.083
Treatment	33	0.606	0.496	0.086

two-sample two-sided t-test

for equal variance

```
1 t.test(school ~ Group,
2       data = pop_dstance_sample_enrollment,
3       var.equal = TRUE)
4 # Two Sample t-test
5 #
6 # data: school by Group
7 # t = -2.6316, df = 62, p-value = 0.01071
8 # alternative hypothesis: true difference in means between group Cont.
9 # 95 percent confidence interval:
10 # -0.55557287 -0.07590318
11 # sample estimates:
12 # mean in group Control mean in group Treatment
13 # 0.2903226 0.6060606
```

for UNEQUAL variance

```
1 t.test(school ~ Group,
2       data = pop_dstance_sample_enrollment,
3       var.equal = FALSE)
4 # Welch Two Sample t-test
5 #
6 # data: school by Group
7 # t = -2.6377, df = 61.995, p-value = 0.01054
8 # alternative hypothesis: true difference in means between group Cont.
9 # 95 percent confidence interval:
10 # -0.55502140 -0.07645465
11 # sample estimates:
12 # mean in group Control mean in group Treatment
13 # 0.2903226 0.6060606
```

- H_0 : School enrollment_T = School enrollment_c
- H_1 : School enrollment_T $\neq$ School enrollment_c
- Conclusion:
 - In both tests $p < 0.05$;
 - School enrollment in treatment is larger than in control at the 5% level of significance. Therefore, the effect of CCT on school enrollment is indeed positive and significant.

Extra materials (for self-study)

t-test

Most commonly used statistical test for hypothesis testing about the
 $\hat{\mu}$

estimated parameter (e.g., mean, but also other estimates).

Its purpose is to infer:

- whether or not the population parameter μ (mean, or else)
- is different from an arbitrary number (usually zero: $\mu_0 = 0$)
- based on our sample estimates of
 $\hat{\mu}$
 - the parameter (mean), its variance (s), and the sample size n .

t-tests

In principal, there are:

- one-sample t-test: compares **one mean to a number**
- two-sample t-test: compares **means of two independent samples**

One-sample t-test: comparing mean to a number

1. **Step 1.** Hypothesis formulation;
2. **Step 2.** Calculating test statistic;
3. **Step 3.** Setting a decision rule for rejecting H_0 ;
4. **Step 4.** Making a conclusion;

Step 1. Hypothesis formulation

H_0 Null Hypothesis

The population mean equals to zero μ_0

- $H_0: \mu = \mu_0$

H_1 Alternative Hypothesis

The population mean is not zero μ_0

- $H_1: \mu \neq \mu_0$
- $H_1: \mu < \mu_0$
- $H_1: \mu > \mu_0$

Step 2. Calculating test statistic

t-statistics:

$$t = \frac{\hat{\mu} - \mu_0}{se} = \frac{\sqrt{n}(\hat{\mu} - \mu_0)}{s},$$

where:

- t - t-statistics;
- $\hat{\mu}$ - estimated mean and μ_0 is an arbitrary number;
- $se = s / \sqrt{n}$ - standard error of the estimated mean;
- s - sample standard deviation;
- n - number of observation;

Step 2. Example

Sample size: $n = 25$

Estimates mean: $\hat{\mu} = -2.2$

Sample SD: $s = 3.3$

Compute the t-statistics for the mean difference from zero.

- $\mu_0 = 0?$

- $t = \frac{\sqrt{25}(-2.2 - 0)}{3.3} = 5 \cdot (-1/3) = -1.667$

Compute the t-statistics for the mean difference from -5.5.

- $\mu_0 = -5.5?$

- $t = \frac{\sqrt{25}(-2.2 + 5.5)}{3.3} = 5 \cdot 1 =$

Step 3. Setting a decision rule (1)

To reject H_0

- we compare t-statistics to a **critical value** $c_{df, CL}$, with
- CL - the confidence level
 - opposite to the significance level: $CL = 1 - \alpha$
- df - degrees of freedom

Step 3. Critical value

is the minimum values of the t-statistics when we can reject H_0

- at the given **confidence level**, and
- with the given **degrees of freedom**.

Step 3. Confidence level

Confidence level depends on the **significance level** (α) and the **alternative hypothesis**:

- when $H_1: \mu \neq \mu_0$ we use a **two-sided** t-test, thus
 - $CL = 1 - \alpha/2$
- when $H_1: \mu < \mu_0$ or $H_1: \mu > \mu_0$ we use a **one-sided** t-test,
 - $CL = 1 - \alpha$

level of significance α ;

- $\alpha = 0.05$

Step 3. Degrees of freedom

the number of degrees of freedom $df = n - k$:

- n is the number of observation in the sample and
- k is the number of parameters
 - $k = 1$ for the one-sample t-test $k = 2$ for the two-sample t-test

Step 3. Reject/Accept

We can reject $H_0: \mu = \mu_0$ in favor of $H_1: \mu \neq \mu_0$ **only if:**

- the absolute value of the t-statistics is strictly greater than the critical value.
- $|t| > c_{df, 1-\alpha/2}$ for two-sided t-test $H_1: \mu \neq \mu_0$
- $|t| > c_{df, 1-\alpha}$ for one-sided t-test $H_1: \mu > \mu_0$ or $H_1: \mu < \mu_0$

In all other cases, we need to accept H_0 .

Step 3. Find a critical value

To find the critical value we use the quantiles of the t-distribution.

- $df = n - k$;
- $CL = 1 - \alpha/2$ for $H_1: \mu \neq \mu_0$
- $CL = 1 - \alpha$ for $H_1: \mu > \mu_0$ or $H_1: \mu < \mu_0$

df	0.9	0.95	0.975	0.99	0.995
1	3.0777	6.3138	12.7062	31.8205	63.6567
5	1.4759	2.0150	2.5706	3.3649	4.0321
10	1.3722	1.8125	2.2281	2.7638	3.1693
25	1.3163	1.7081	2.0595	2.4851	2.7874
40	1.3031	1.6839	2.0211	2.4233	2.7045
50	1.2987	1.6759	2.0086	2.4033	2.6778
75	1.2929	1.6654	1.9921	2.3771	2.6430
100	1.2901	1.6602	1.9840	2.3642	2.6259
150	1.2872	1.6551	1.9759	2.3515	2.6090
300	1.2844	1.6499	1.9679	2.3388	2.5923
1000	1.2824	1.6464	1.9623	2.3301	2.5808
10000	1.2816	1.6450	1.9602	2.3267	2.5763
1000000	1.2816	1.6449	1.9600	2.3264	2.5758

Step 3. Example (1)

Sample size: $n = 25$; $df = 25 - 1 = 24$

Is the population mean different from zero at a 1% level of significance?

- $H_1: \mu \neq 0$ - two-sided t-test ($\alpha/2$)
- $1 - \alpha/2 = 1 - 0.01/2 = 0.995$
- $c_{24, 1-0.01/2} = 2.7969$

df	0.95	0.975	0.99	0.995
1	6.3138	12.7062	31.8205	63.6567
5	2.0150	2.5706	3.3649	4.0321
10	1.8125	2.2281	2.7638	3.1693
20	1.7247	2.0860	2.5280	2.8453
23	1.7139	2.0687	2.4999	2.8073
24	1.7109	2.0639	2.4922	2.7969
25	1.7081	2.0595	2.4851	2.7874
26	1.7056	2.0555	2.4786	2.7787
27	1.7033	2.0518	2.4727	2.7707
28	1.7011	2.0484	2.4671	2.7633
29	1.6991	2.0452	2.4620	2.7564
30	1.6973	2.0423	2.4573	2.7500
50	1.6759	2.0086	2.4033	2.6778
150	1.6551	1.9759	2.3515	2.6090

Step 3. Example (2)

Sample size: $n = 25$; $df = 25 - 1 = 24$

Is the population mean different from zero at a 5% level of significance?

- $H_1: \mu \neq 0$ - two-sided t-test ($\alpha/2$)
- $1 - \alpha/2 = 1 - 0.05/2 = 0.975$
- $c_{24, 1-0.05/2} = 2.0639$

df	0.95	0.975	0.99	0.995
1	6.3138	12.7062	31.8205	63.6567
5	2.0150	2.5706	3.3649	4.0321
10	1.8125	2.2281	2.7638	3.1693
20	1.7247	2.0860	2.5280	2.8453
23	1.7139	2.0687	2.4999	2.8073
24	1.7109	2.0639	2.4922	2.7969
25	1.7081	2.0595	2.4851	2.7874
26	1.7056	2.0555	2.4786	2.7787
27	1.7033	2.0518	2.4727	2.7707
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29	1.6991	2.0452	2.4620	2.7564
30	1.6973	2.0423	2.4573	2.7500
50	1.6759	2.0086	2.4033	2.6778
150	1.6551	1.9759	2.3515	2.6090

Step 3. Example (3)

Sample size: $n = 25$; $df = 25 - 1 = 24$.

Is the population mean different from zero at 10% level of significance?

- $H_1: \mu \neq 0$ - two-sided t-test ($\alpha/2$)
- $1 - \alpha/2 = 1 - 0.1/2 = 0.95$
- $c_{24, 1-0.1/2} = 1.7109$

df	0.95	0.975	0.99	0.995
1	6.3138	12.7062	31.8205	63.6567
5	2.0150	2.5706	3.3649	4.0321
10	1.8125	2.2281	2.7638	3.1693
20	1.7247	2.0860	2.5280	2.8453
23	1.7139	2.0687	2.4999	2.8073
24	1.7109	2.0639	2.4922	2.7969
25	1.7081	2.0595	2.4851	2.7874
26	1.7056	2.0555	2.4786	2.7787
27	1.7033	2.0518	2.4727	2.7707
28	1.7011	2.0484	2.4671	2.7633
29	1.6991	2.0452	2.4620	2.7564
30	1.6973	2.0423	2.4573	2.7500
50	1.6759	2.0086	2.4033	2.6778
150	1.6551	1.9759	2.3515	2.6090

Step 4. Make a conclusion

Concluding either or:

- We reject H_0
 - at least at the α level of significance.
 - the population mean is significantly different from μ_0 .
- We fail to reject (we accept) H_0 .
 - available data do not provide sufficient evidence to reject H_0 at the α level of significance.
 - the population mean is statistically indifferent from μ_0 .

Step 4. Example (1)

Sample size: $n = 25$. Estimates mean: $\hat{\mu} = -2.2$. Sample SD: $s = 3.3$

Is the population mean different from zero ($\mu_0 = 0$) at 5% level of significance?

- $t = \frac{\sqrt{25}(-2.2-0)}{3.3} = -1.667$
- $c_{24, 1-0.05/2} = 2.0639$
- $|t| < c_{24, 1-0.05/2}$
- $|-1.667| < 2.0639$
- Conclusion:
 - Fail to reject (we accept) H_0 .
 - the population mean is statistically **indifferent** from $\mu_0 = 0$.

Step 4. Example (2)

Is the population mean different from -5.5 at the 1% level of significance?

- $\mu_0 = -5.5$?
- $t = \frac{\sqrt{25}(-2.2 + 5.5)}{3.3} = 5$
- $c_{24, 1-0.05/2} = 2.7969$
- $|t| > c_{24, 1-0.01/2}$
- $|5| > 2.7969$
- Conclusion:
 - Reject H_0 .
 - the population mean is statistically different from $\mu_0 = -5.5$ at least at the 1% level of significance.

Examples (1)

Calculate the t-statistics and find corresponding critical values.

- Conclude whether the H_0 could be rejected in the following hypothesis testing problems.
- Compare your calculations to the solutions.

$\hat{\mu}$	s	μ_0	n	α	t	$c_{df, CL}$
4.0	1.0	3.0	10	0.01	3.162	3.169
2.0	1.2	0.8	8	0.05	2.828	2.306
3.5	3.1	2.8	150	0.01	2.766	2.609
8.0	4.0	16.0	21	0.10	-9.165	1.721
1.2	3.4	0.2	53	0.05	2.141	2.006

Critical values of the t-distribution.

df	0.95	0.975	0.99	0.995
1	6.3138	12.7062	31.8205	63.6567
5	2.0150	2.5706	3.3649	4.0321
6	1.9432	2.4469	3.1427	3.7074
7	1.8946	2.3646	2.9980	3.4995
8	1.8595	2.3060	2.8965	3.3554
9	1.8331	2.2622	2.8214	3.2498
10	1.8125	2.2281	2.7638	3.1693
20	1.7247	2.0860	2.5280	2.8453
25	1.7081	2.0595	2.4851	2.7874
30	1.6973	2.0423	2.4573	2.7500
50	1.6759	2.0086	2.4033	2.6778
80	1.6641	1.9901	2.3739	2.6387
100	1.6602	1.9840	2.3642	2.6259
150	1.6551	1.9759	2.3515	2.6090

t-scores

The slides below contain the t-scores (critical values of the t-distribution) for t-distributions with 1-150 degrees of freedom.

t-scores (1)

df	0.9	0.95	0.975	0.99	0.995
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947

df	0.9	0.95	0.975	0.99	0.995
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750

t-scores (2)

df	0.9	0.95	0.975	0.99	0.995
31	1.309	1.696	2.040	2.453	2.744
32	1.309	1.694	2.037	2.449	2.738
33	1.308	1.692	2.035	2.445	2.733
34	1.307	1.691	2.032	2.441	2.728
35	1.306	1.690	2.030	2.438	2.724
36	1.306	1.688	2.028	2.434	2.719
37	1.305	1.687	2.026	2.431	2.715
38	1.304	1.686	2.024	2.429	2.712
39	1.304	1.685	2.023	2.426	2.708
40	1.303	1.684	2.021	2.423	2.704
41	1.303	1.683	2.020	2.421	2.701
42	1.302	1.682	2.018	2.418	2.698
43	1.302	1.681	2.017	2.416	2.695
44	1.301	1.680	2.015	2.414	2.692
45	1.301	1.679	2.014	2.412	2.690

df	0.9	0.95	0.975	0.99	0.995
46	1.300	1.679	2.013	2.410	2.687
47	1.300	1.678	2.012	2.408	2.685
48	1.299	1.677	2.011	2.407	2.682
49	1.299	1.677	2.010	2.405	2.680
50	1.299	1.676	2.009	2.403	2.678
51	1.298	1.675	2.008	2.402	2.676
52	1.298	1.675	2.007	2.400	2.674
53	1.298	1.674	2.006	2.399	2.672
54	1.297	1.674	2.005	2.397	2.670
55	1.297	1.673	2.004	2.396	2.668
56	1.297	1.673	2.003	2.395	2.667
57	1.297	1.672	2.002	2.394	2.665
58	1.296	1.672	2.002	2.392	2.663
59	1.296	1.671	2.001	2.391	2.662
60	1.296	1.671	2.000	2.390	2.660

t-scores (3)

df	0.9	0.95	0.975	0.99	0.995
65	1.295	1.669	1.997	2.385	2.654
70	1.294	1.667	1.994	2.381	2.648
75	1.293	1.665	1.992	2.377	2.643
80	1.292	1.664	1.990	2.374	2.639
85	1.292	1.663	1.988	2.371	2.635
90	1.291	1.662	1.987	2.368	2.632
95	1.291	1.661	1.985	2.366	2.629
100	1.290	1.660	1.984	2.364	2.626
105	1.290	1.659	1.983	2.362	2.623
110	1.289	1.659	1.982	2.361	2.621
115	1.289	1.658	1.981	2.359	2.619
120	1.289	1.658	1.980	2.358	2.617
125	1.288	1.657	1.979	2.357	2.616
130	1.288	1.657	1.978	2.355	2.614
135	1.288	1.656	1.978	2.354	2.613

df	0.9	0.95	0.975	0.99	0.995
140	1.288	1.656	1.977	2.353	2.611
145	1.287	1.655	1.976	2.352	2.610
150	1.287	1.655	1.976	2.351	2.609
155	1.287	1.655	1.975	2.351	2.608
160	1.287	1.654	1.975	2.350	2.607
165	1.287	1.654	1.974	2.349	2.606
170	1.287	1.654	1.974	2.348	2.605
175	1.286	1.654	1.974	2.348	2.604
180	1.286	1.653	1.973	2.347	2.603
185	1.286	1.653	1.973	2.347	2.603
190	1.286	1.653	1.973	2.346	2.602
195	1.286	1.653	1.972	2.346	2.601
200	1.286	1.653	1.972	2.345	2.601
250	1.285	1.651	1.969	2.341	2.596
300	1.284	1.650	1.968	2.339	2.592

t-scores (4)

df	0.9	0.95	0.975	0.99	0.995
300	1.284	1.650	1.968	2.339	2.592
350	1.284	1.649	1.967	2.337	2.590
400	1.284	1.649	1.966	2.336	2.588
450	1.283	1.648	1.965	2.335	2.587
500	1.283	1.648	1.965	2.334	2.586
550	1.283	1.648	1.964	2.333	2.585
600	1.283	1.647	1.964	2.333	2.584
650	1.283	1.647	1.964	2.332	2.583
700	1.283	1.647	1.963	2.332	2.583
750	1.283	1.647	1.963	2.331	2.582
800	1.283	1.647	1.963	2.331	2.582
850	1.283	1.647	1.963	2.331	2.582
900	1.282	1.647	1.963	2.330	2.581
950	1.282	1.646	1.962	2.330	2.581
1000	1.282	1.646	1.962	2.330	2.581

df	0.9	0.95	0.975	0.99	0.995
2000	1.282	1.646	1.961	2.328	2.578
3000	1.282	1.645	1.961	2.328	2.577
4000	1.282	1.645	1.961	2.327	2.577
5000	1.282	1.645	1.960	2.327	2.577
6000	1.282	1.645	1.960	2.327	2.577
7000	1.282	1.645	1.960	2.327	2.577
8000	1.282	1.645	1.960	2.327	2.576
9000	1.282	1.645	1.960	2.327	2.576
10000	1.282	1.645	1.960	2.327	2.576

Homework

Homework

This homework generally outlines examples of the exam questions.

1. Use paper ([Duflo, Kremer, and Robinson 2008](#)) (available on Ilias);
 - Perform a statistical hypothesis testing of the mean yield increase for panel A-C in table 1.
 - State H_0 and H_1 ;
 - Conclude about the effect of fertilizer application on the yields for each of the panels.
2. Watch “[Randomized Trials: The Ideal Weapon](#)” (It was suggested already in one of the previous lectures)
 - Take the paper ([Carter, Greenberg, and Walker 2017](#)) (available on Ilias) and interpret results of the columns 4-5 in the table 2.

Key concepts covered in the lecture

T-distribution:

- Normal vs t-distribution.
- t-scores;
- Identifying t-scores from a table;

Confidence intervals:

- For the population observations and normal distribution;
- For the sample estimates of mean.
- Construction of the confidence intervals.

t-test:

- two-sided t-test;
- Hypotheses null and alternative;
- Formulating Hypotheses;
- Rejecting/accepting:
 - test of significance; p-value; confidence intervals;
- Interpreting t-test results;
- Concluding about the mean difference based on the t-test;

References

- Angrist, Joshua D., and Jörn-Steffen Pischke. 2014. *Mastering 'metrics: The Path from Cause to Effect*. Princeton University Press.
- Carter, Susan Payne, Kyle Greenberg, and Michael S. Walker. 2017. "The Impact of Computer Usage on Academic Performance: Evidence from a Randomized Trial at the United States Military Academy." *Economics of Education Review* 56 (February): 118–32. <https://doi.org/10.1016/j.econedurev.2016.12.005>.
- Diez, David, Mine Cetinkaya-Rundel, and Christopher D Barr. 2022. *OpenIntro Statistics: Fourth Edition*. OpenIntro. <https://www.openintro.org/book/os/>.
- Duflo, Esther, Michael Kremer, and Jonathan Robinson. 2008. "How High Are Rates of Return to Fertilizer? Evidence from Field Experiments in Kenya." *American Economic Review* 98 (2): 482–88. <https://doi.org/10.1257/aer.98.2.482>.
- Fiszbein, Ariel, Norbert Schady, Francisco H. G. Ferreira, Margaret Grosh, Niall Keleher, Pedro Olinto, and Emmanuel Skoufias. 2009. *Conditional Cash Transfers : Reducing Present and Future Poverty*. The World Bank. <https://doi.org/10.1596/978-0-8213-7352-1>.
- Gertler, Paul J., Sebastian Martinez, Patrick Premand, Laura B. Rawlings, and Christel M. J. Vermeersch. 2016. *Impact Evaluation in Practice, Second Edition*. Washington, DC: Inter-American Development Bank; World Bank. <https://doi.org/10.1596/978-1-4648-0779-4>.
- Miguel, Edward, and Michael Kremer. 2004. "Worms: Identifying Impacts on Education and Health in the Presence of Treatment Externalities." *Econometrica* 72 (1): 159–217. <https://doi.org/10.1111/j.1468-0262.2004.00481.x>.